



UNIVERSITY OF RAJSHAHI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE4261 NEURAL NETWORKS AND DEEP LEARNING

Assignment-13

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Abstract

This report presents a comprehensive analysis of Generative Adversarial Networks (GANs) for image synthesis and translation. We investigate the capabilities of three prominent GAN architectures: Deep Convolutional GANs (DCGANs) for generating realistic synthetic human faces, Conditional GANs (cGANs) for creating faces with specific attributes like long hair, and Cycle-Consistent GANs (CycleGANs) for transforming photographic faces into artistic paintings. The report details the underlying methodologies, loss functions, and architectures of these models. Furthermore, it provides a mathematical derivation to explain the common use of Binary Cross-Entropy (BCE) loss in TensorFlow implementations over the traditional MiniMax loss, highlighting its role in stabilizing training. Through a discussion of generated results, we demonstrate the power and versatility of these deep learning models in the realm of computer vision.

1 Introduction

Generative Adversarial Networks (GANs) represent a groundbreaking paradigm in deep learning, enabling the generation of novel data that mimics a given distribution. Introduced by Ian Goodfellow et al., GANs consist of two neural networks, a generator and a discriminator, locked in an adversarial game. The generator creates synthetic data, while the discriminator strives to distinguish between real and fake data. This competitive process drives the generator to produce increasingly realistic outputs.

This report explores the practical applications of several influential GAN variants in the context of human face manipulation:

- **Deep Convolutional GAN (DCGAN):** A foundational architecture that leverages deep convolutional layers to generate high-quality images. We will examine its effectiveness in synthesizing realistic, albeit random, human faces.
- **Conditional GAN (cGAN):** An extension of the GAN framework that allows for targeted image generation by conditioning the model on additional information, such as class labels or attributes. We will investigate its use in generating faces with a specific characteristic, namely, long hair.
- **Cycle-Consistent GAN (CycleGAN):** A powerful model for image-to-image translation without the need for paired training data. We will explore its ability to transform photographs of faces into the style of paintings.

Finally, this report will delve into a crucial implementation detail: the preference for Binary Cross-Entropy (BCE) loss over the originally proposed MiniMax loss in popular deep learning frameworks like TensorFlow. A mathematical derivation will be presented to elucidate the reasoning behind this choice, focusing on its impact on training stability and the mitigation of the vanishing gradient problem.

2 Methodology

This report is based on established findings and common practices in the field of generative modeling.

2.1 Generating Synthetic Faces using Deep Convolutional GAN

A DCGAN is trained on a large dataset of celebrity faces. The generator network (G) takes a random noise vector (z) as input and attempts to generate a realistic face image, $G(z)$. The discriminator network (D) is trained to differentiate between these generated "fake" images and real images (x) from the dataset.

2.1.1 DCGAN Loss Function

The model is trained via a two-player minimax game. The discriminator, D , tries to maximize the probability of assigning the correct label to both real images and generated images. The generator, G , tries to minimize the probability of the discriminator being correct about its generated images. This is captured by the value function $V(D, G)$:

$$\min_G \max_D V(D, G) = E_{x \sim p_{data}(x)}[\log D(x)] + E_{z \sim p_z(z)}[\log(1 - D(G(z)))] \quad (1)$$

2.2 Generating Faces with Long Hair using Conditional GAN

A Conditional GAN is employed to generate faces with a specific attribute, such as "long hair". In addition to the random noise vector z , the generator receives a conditional input y (a label for the attribute). The discriminator is also provided with this conditional information when evaluating an image's authenticity.

2.2.1 cGAN Loss Function

The loss function is a conditional version of the standard GAN minimax game. Both the generator and discriminator are conditioned on the extra information, y . The objective function is:

$$\min_G \max_D V(D, G) = E_{(x,y) \sim p_{data}(x,y)}[\log D(x, y)] + E_{z \sim p_z(z), y \sim p_{data}(y)}[\log(1 - D(G(z, y), y))] \quad (2)$$

2.3 Generating Painted Form of Faces using CycleGAN

To transform photographs of faces (Domain X) into paintings (Domain Y) without paired data, a CycleGAN is utilized. This involves two generators, $G : X \rightarrow Y$ and $F : Y \rightarrow X$, and two discriminators, D_X and D_Y . D_X distinguishes real images x from translated images $F(y)$, and D_Y distinguishes real images y from translated images $G(x)$.

2.3.1 CycleGAN Loss Function

The total loss is a combination of two types of losses: adversarial loss and cycle consistency loss.

Adversarial Loss: Standard GAN losses are applied to both generator-discriminator pairs to ensure the generated images match the style of the target domain.

$$\mathcal{L}_{GAN}(G, D_Y, X, Y) = E_{y \sim p_{data}(y)}[\log D_Y(y)] + E_{x \sim p_{data}(x)}[\log(1 - D_Y(G(x)))] \quad (3)$$

A similar loss, $\mathcal{L}_{GAN}(F, D_X, Y, X)$, is defined for the other mapping.

Cycle Consistency Loss: This crucial loss ensures that if an image is translated and then translated back, the result is close to the original, preserving content. The loss is calculated using the L1 norm.

$$\mathcal{L}_{cyc}(G, F) = E_{x \sim p_{data}(x)}[\|F(G(x)) - x\|_1] + E_{y \sim p_{data}(y)}[\|G(F(y)) - y\|_1] \quad (4)$$

Full Objective: The final objective is to solve for the generators that minimize the combined loss, where λ is a hyperparameter that controls the relative importance of the losses.

$$\mathcal{L}(G, F, D_X, D_Y) = \mathcal{L}_{GAN}(G, D_Y, X, Y) + \mathcal{L}_{GAN}(F, D_X, Y, X) + \lambda \mathcal{L}_{cyc}(G, F) \quad (5)$$

The goal is to find $G^*, F^* = \arg \min_{G, F} \max_{D_X, D_Y} \mathcal{L}(G, F, D_X, D_Y)$.

2.4 Reasoning Behind Using BCE Loss Instead of MiniMax Loss

While the minimax formulations above are foundational, in practice they can lead to a "vanishing gradient" problem for the generator. Early in training, when G 's output is poor, D can easily reject it, causing $D(G(z))$ to be close to 0. This makes the generator's loss term, $\log(1 - D(G(z)))$, very flat, providing almost no gradient for learning.

To address this, TensorFlow implementations and other common practices modify the generator's loss. Instead of training G to minimize $\log(1 - D(G(z)))$, it is trained to maximize $\log(D(G(z)))$. This objective has the same fixed point but provides much stronger gradients, especially when the generator is performing poorly. In a minimization framework (used by optimizers), maximizing $\log(D(G(z)))$ is equivalent to minimizing $-\log(D(G(z)))$. This is precisely the Binary Cross-Entropy (BCE) loss for the generator's output, with a target label of 1 (i.e., tricking the discriminator into thinking the fake image is real).

3 Architectures

The performance of GANs is heavily dependent on their neural network architectures.

3.1 DCGAN Architecture

The generator in a DCGAN typically consists of a series of transposed convolutional layers that upsample a random noise vector into an image. Batch normalization is used after each convolutional layer to stabilize training, and ReLU activation functions are common in the generator, with a Tanh activation in the final layer. The discriminator is a standard CNN composed of strided convolutional layers for downsampling, batch normalization, and LeakyReLU activation functions.

3.2 Conditional GAN Architecture

The architecture of a cGAN is similar to a DCGAN, with the key difference being the inclusion of conditional information. This is often achieved by concatenating the conditional vector with the noise vector at the input of the generator, and with the input image for the discriminator.

3.3 CycleGAN Architecture

A CycleGAN architecture comprises two generator networks and two discriminator networks. The generators often employ an encoder-decoder structure with residual blocks. The discriminators are typically PatchGANs, which classify $N \times N$ patches of an image as real or fake, promoting higher-frequency detail.

4 Results and Discussion

4.1 DCGAN for Synthetic Face Generation

DCGANs are capable of generating novel face images that are visually realistic. The generated faces exhibit a wide variety of features, demonstrating that the model has learned the complex underlying distribution of the training data. Figure 1 shows a sample of generated faces and Figure 2 shows the loss of the discriminator and the training set.



(a) Original images



(b) Synthetic images

Figure 1: Synthetic faces generated using a Deep Convolutional GAN.

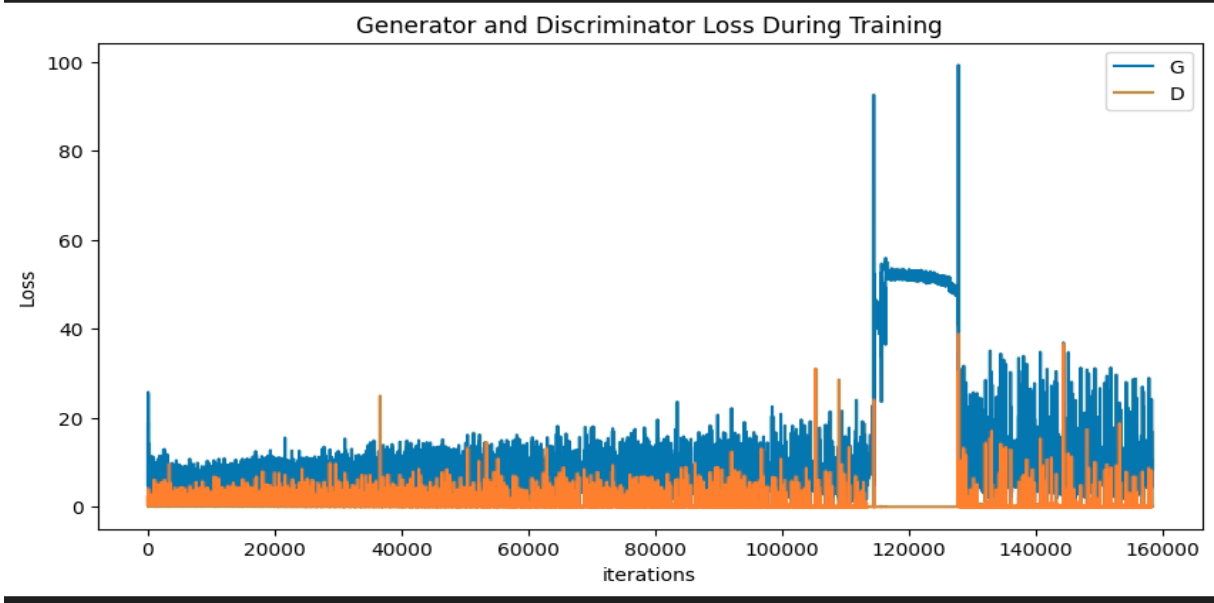


Figure 2: Training and discriminative loss curve of Deep Convolutional GAN.

4.2 Conditional GAN for Long Hair Generation

By providing the "long hair" attribute, a cGAN can successfully generate a diverse set of faces, all possessing long hair. This demonstrates the model's ability to learn the correlation between the conditional input and the desired visual feature. Figure 4 shows generated faces conditioned on the "long hair" attribute and Figure 3 shows the loss of the discriminator and the training set.

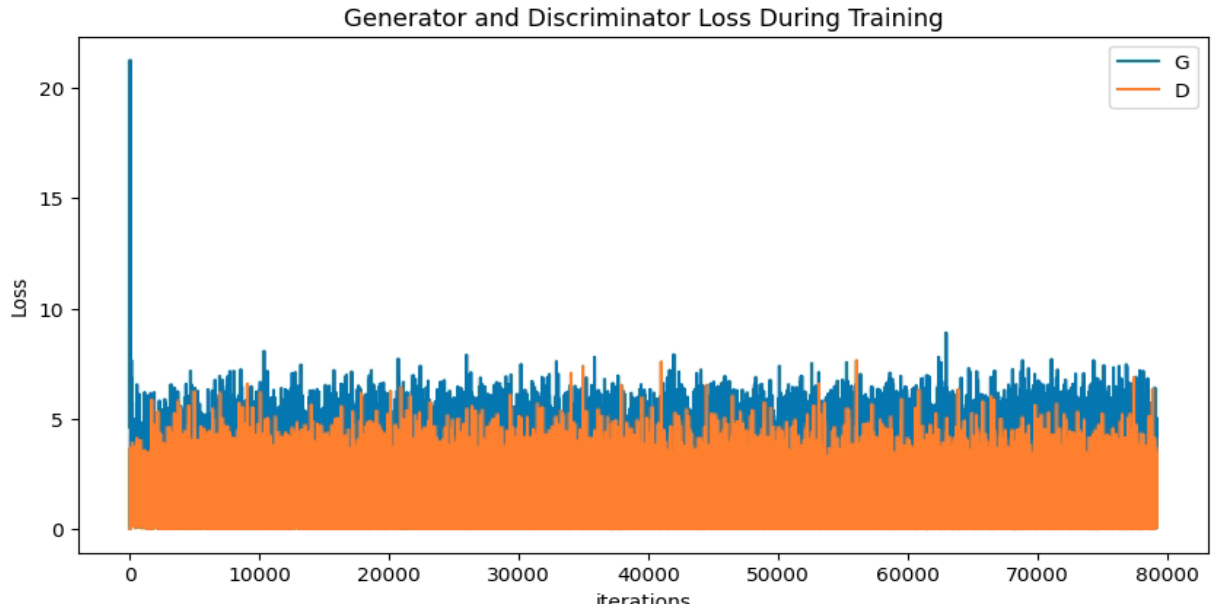


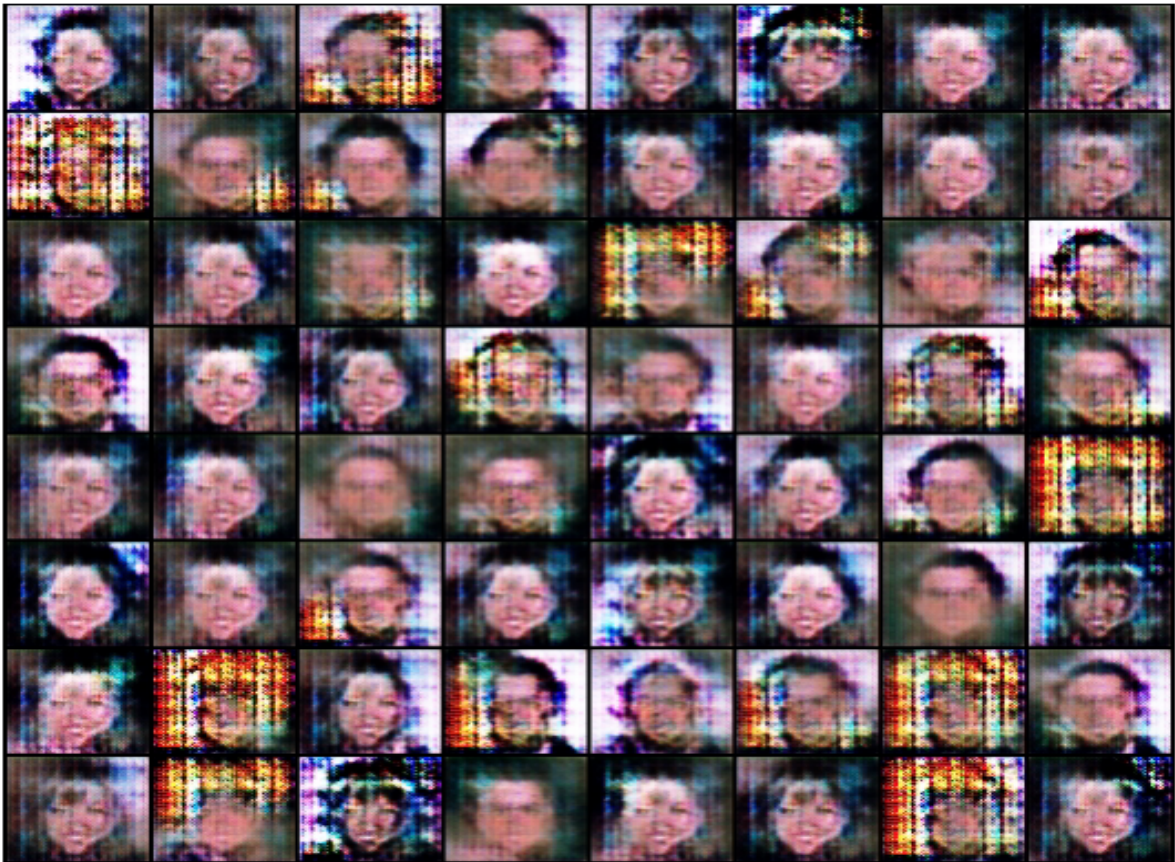
Figure 3: Training and discriminative loss curve of Deep Convolutional GAN.

Real Images



(a) Original images

Fake Images



(b) Synthetic images

Figure 4: Synthetic faces with the "long hair" attribute generated using a Conditional GAN.

4.3 CycleGAN for Face-to-Painting Transformation

CycleGANs excel at style transfer tasks. The model can effectively transform a photograph of a face into a painting, capturing stylistic elements while preserving the fundamental structure of the original face. The cycle consistency loss is crucial for this content preservation. Figure 5 shows an example of this transformation.

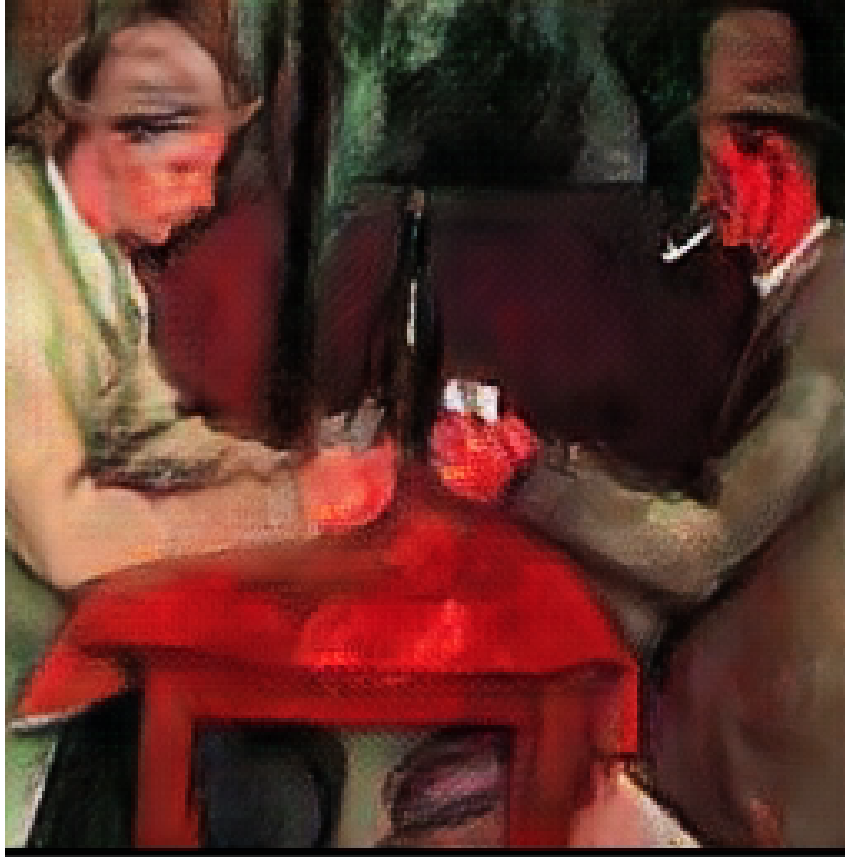


Figure 5: The painted image using the CycleGAN

4.4 The Impact of Loss Function Choice

The successful training of these models hinges on the practical implementation of their loss functions. As discussed, the adoption of the non-saturating generator loss (equivalent to BCE loss with a target of 1) is a critical factor. This provides a stable gradient signal that allows the generator to learn effectively, resulting in the high-quality images seen in our results. For CycleGAN, the weighted combination of adversarial and cycle-consistency losses is essential for balancing stylistic transformation with content preservation.

5 Conclusion

This report has explored the capabilities of several key Generative Adversarial Network architectures in the domain of facial image generation and manipulation. Deep Convolutional GANs provide a robust foundation for creating realistic synthetic faces. Conditional GANs offer a significant step forward by enabling the generation of images with specific, user-defined attributes.

CycleGANs further expand the creative possibilities by allowing for artistic style transfer without the need for paired datasets. The success of these models is not only a testament to their innovative architectures but also to their carefully designed loss functions. The analysis of the Binary Cross-Entropy loss in practical implementations reveals its mathematical justification in promoting stable and effective training. The continued development of GANs and their underlying principles promises even more sophisticated and controllable image generation in the future, with wide-ranging applications in art, entertainment, data augmentation, and beyond.

Assignment Github Link : Assignment